

24 GHz FMCW-RADAR KIT

OVERVIEW

The **24 GHz 2-Channel FMCW-Radar** is a miniaturized, ultralight radar unit. This system comprises a manifold configurable RF fronted with digital signal processing and programmable interface. The controller allows flexible generation of radar signals and provides enough computation power to perform signal processing for target detection or even complex tracking algorithms.

The radar unit is equipped with Ethernet interface which allows the user to define all radar settings and to receive all measured data. These data can be calculated properties from single targets (like distance, angle and speed) or complex time- or frequency domain curves for user-defined evaluations.

The **Developer Kit** comprises the Radar Sensor, a PC software with Graphical User Interface and a LAN cable for user-friendly startup, testing and application specific integration of the radar unit. The software allows parameter settings and visualization of measured radar data. Typical parameters are detection range, resolution, frequency bandwidth, output power, radar



signal shape and duration. The measurement results are shown in different types of graphs and can be exported to files in text or binary format.

An **Expert Mode** allows the experienced user to test the radar sensor with modified settings like frequency bandwidth or RF output power.

FEATURES:

- 24 GHz ISM-Band Radar-Module appl. FMCW
- 2 Receive Antennas with I/Q Channels each
- Distance and Velocity Measurements
- Target Detection with Angle Information
- Onboard Signal Processing with DSC
- Ethernet (Power over Ethernet) Interface
- Weatherproofed Housing
- PC-Software with Graphical User Interface

APPLICATIONS:

- Human Tracking, Intruder Detection
- Obstacle Recognition
- Distance and Speed Monitoring
- Terrain Mapping, Awareness and Warning (TAWS)
- Dedicated for UAVs and Drones

TECHNICAL DATA sR-1200e

GENERAL

Modulation:	FMCW / CW
Operating Frequency:	24.0GHz - 24.25GHz (ISM band)
Number of Channels:	1 Tx, 2 Rx (with I/Q demodulator each)
Data Interface:	Ethernet (with PoE)
Certification:	CE-Approval

ANTENNA

Antenna Type:	Integrated Patch Antenna
Antenna Characteristics:	65° Azimuth, 24° Elevation
Antenna Polarization:	linear

MEASUREMENT

Min. Measurement Range:	0.6m (@ ISM band)
Max. Measurement Range:	307m (@ ISM band)
Range Resolution:	max. 0.6m (@ ISM band)
Min. Measurement Speed:	-3200m/s (@ 24GHz, CW Mode)
Max. Measurement Speed:	3200m/s (@ 24 GHz, CW Mode)
Speed Resolution:	6.25m/s (@ 24GHz, CW Mode)
Angle Measurement Accuracy:	2° - 3°

FMCW PERFORMANCE

Frequency Ramp Duration:	1ms - 100ms
Typical Update Rate:	10Hz - 200Hz (depending on application)
Output Power (EIRP):	10dBm - 19dBm (tunable)

TEMPERATURE

Min. Operating Temperature:	-40°C
Max. Operating temperature:	60°C (@ duty cycle < 50%)

POWER SUPPLY

Operation Voltage:	44V - 54V (PoE)
Standby Power:	3.0W
Operating Power:	4.5W

HOUSING

Dimensions (L x W x H):	98mm x 87mm x 42.5mm (Housing) 114mm x 87mm x 42.5mm (with Bushing)
Weight:	280g (with cable)
Mounting:	4 Mounting Holes (5mm)
Connection Cable and Connector:	CAT-6 with RJ45 Female Plug (Ethernet)
Protection Code for Housing:	IP65



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